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* when the universe of data is Unknown

↳ Discrimination models (Classification / Regression)

→ Principle :

if u is near / similar to u' then

$f(u)$ should be near $f(u')$

↓

Not always True !

① $f(u)$ is a class $c \in \{1, 2, \dots, n\}$

$$f(u) = c = f(u')$$

② Regression $|f(u) - f(u'|$

→ How to calculate u near u' : (vectors in $\mathbb{R}^b$)

1. $\|u\| \geq 0$
2. $\|\alpha u\| = |\alpha| \cdot \|u\|$
3. $\|u + u'\| \leq \|u\| + \|u'\|$

$\Big\}$ $d = \|u - u'\|$

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* Types of Norms :

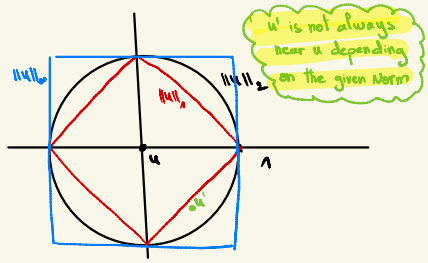
$$\|u\|_2 = \sqrt{\sum_{k=1}^{d} u_k^2}$$

$$\|u\|_1 = \sum_{k=1}^{d} |u_k|$$

$$\|u\|_\infty = \max_{k=1,\dots,d} \{|u_k|\}$$

} Are equivalent At the Limit ⇒

example : u is near u' if $d(u,u') \leq 1$



u' is not always near u depending on the given Norm

* We can't compare data points with different units :

→ We Normalize :

* 2-score : $\hat{u}_1 = \frac{u_1 - \mu_1}{\sigma_1}$ (μ_1 → mean, σ_1 → std)

$$\mu_1 = \frac{1}{m} \sum_{k=1}^{m} u_1(k)$$

$$\sigma_1 = \sqrt{\frac{1}{m} \sum_{k=1}^{m} (u_1(k) - \mu_1)^2}$$

$\sigma_1 = 0 \Rightarrow u_1 \equiv \mu_1$

Unitless ! ⊕ Lossless info

* Another one : $\hat{u}_1 = \frac{u_1 - \min(u_1)}{\max(u_1) - \min(u_1)}$

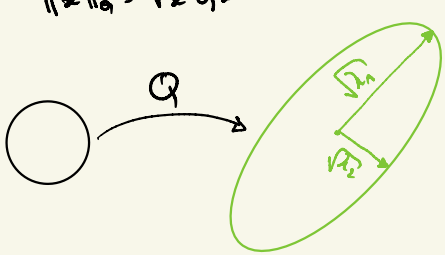
* Quadratic Norms :

- Let Q a.s.d positive definite matrix ($\forall x, x^T Q x \geq 0$)

$$\|x\|_Q = \sqrt{x^T Q x}$$

$Q = R \Lambda R^T$ ($R.R^T = I$)

↓ Rotation Matrix ($\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$)



∴ Q stretches the space.

- $Q = I \iff$ Minkowsky $p = 2$

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* why is $Q = I$ a ☆ :

→ Dot product : $x \cdot y = \sum_{i=1}^{d} (x_i \cdot y_i)$ $(x, y \in \mathbb{R}^d)$ → $x^T y$

$\|x\|_2^2 = x_1^2 + \dots + x_d^2 = x \cdot x$

$x \cdot y = \|x\| \cdot \|y\| \cdot \cos(\hat{x,y}) \rightarrow |x \cdot y| \leq \|x\| \ \|y\|$

Cauchy - schwartz

- $\|x\|_Q^2 = x^T Q x = x^T (R \Lambda R^T) x = (R^T x)^T \Lambda (R^T x)$

$\lambda_i > 0$: $(\sqrt{\Lambda}\, R^T x)^T (\sqrt{\Lambda}\, R^T x) = X^T X = X \cdot X$

X

the euclidian Norm Here is a dot product there

- std — is dot product :

$\sigma_u^2 = \frac{1}{m} \sum_{k=1}^{m} (u(k) - \mu_u)^2$ $\quad X = \frac{1}{\sqrt{m}} (u - \mu_u)$

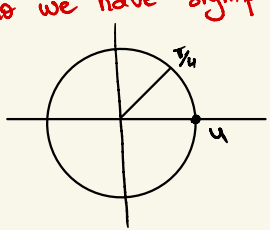
$\hookrightarrow \sigma_u^2 = X \cdot X$

- Correlation :

$$Corr(u,v) = \frac{\frac{1}{m} \sum (u_k - \mu_u)(v_k - \mu_v)}{\sqrt{\frac{1}{m} \sum (u_k - \mu_u)^2} \cdot \sqrt{\frac{1}{m} \sum (v_k - \mu_v)^2}} = \frac{\tilde{u} \cdot \tilde{v}}{\|\tilde{u}\| \cdot \|\tilde{v}\|} = \cos(\hat{u,v})$$

$$\begin{cases} \tilde{u} = u_k - \mu_u \\ \tilde{v} = v_k - \mu_v \end{cases}$$

* when do we have significant correlation ?



$\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} = 0.7071$

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Exercice:

- Draw 1000 $U, V \in \mathbb{R}^{100}$ uniform random

- $C_k = \text{Corr}(U, V)$ $k = 1, \ldots, 1000$.

- Plot Histogram of C_k.

Q: Does correlation implies causality ?

↳ NO !

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